

Understanding
functioning and
the brain
through...

FACIAL ANALYSIS

WHAT IS THIS PROJECT?

We study how the **brain controls facial movements and speech**. By carefully **tracking** how your face moves with cameras, we can learn which parts of the brain are responsible for these functions.



WHAT HAPPENS DURING AWAKE BRAIN SURGERY?

In some **brain surgeries**, patients are awake for a short time. This helps surgeons protect important functions like: ***speaking, moving the face,*** and ***understanding language.***

During surgery, patients are asked to do simple tasks, such as talking or making facial movements.

WHY ARE FACIAL EXPRESSIONS IMPORTANT?

Your **face** is directly controlled by the **brain**. Even very small changes in facial movement can show if a certain brain area is affected.

Right now, doctors mainly rely on what they can see and hear. But some changes are too **subtle** for the human eye. And surgery comes with many **time constraints**.

OUR GOAL

Our goal is to support **surgeons** and **clinicians** during **awake brain surgery** with **facial analysis**:

- giving quantitative, measurable, retro and real-time feedback
- helping detect early signs of speech or facial problems
- supporting safety while removing brain tumors

